

ATKA, Russian Federation

WMO: 259020

Lat: 60.850N

Lon: 151.767E

Elev: 901

StdP: 90.96

Time Zone: 11.00 (E11)

Period: 86-95

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.2	-39.2	-43.1	0.1	-40.9	-41.3	0.1	-39.1	12.2	-29.0	11.1	-30.7	0.3	0	0.886

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	23.7	14.1	21.7	13.0	19.6	12.0	14.7	22.1	13.6	20.6	12.5	18.7	3.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.4	9.3	17.4	10.2	8.6	16.3	9.1	8.0	14.8	44.4	22.2	41.0	20.7	38.1	18.7	18.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.7	8.7	7.8	-42.8	26.9	3.0	1.7	-45.0	28.1	-46.7	29.0	-48.4	30.0	-50.6	31.2		
(5)			-42.8	16.4	2.8	1.6	-44.8	17.5	-46.4	18.4	-47.9	19.3	-50.0	20.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-9.7	-30.0	-26.7	-20.8	-11.6	-0.7	9.5	12.5	9.0	1.8	-9.8	-21.8	-28.6
	DBStd	16.07	5.03	5.36	6.29	5.09	4.11	4.54	3.45	4.16	4.43	6.33	6.73	6.69
	HDD10.0	7365	1241	1028	954	648	332	64	16	68	246	615	955	1198
	HDD18.3	10228	1499	1262	1212	898	591	264	182	290	496	873	1205	1456
	CDD10.0	181	0	0	0	0	0	51	93	37	1	0	0	0
	CDD18.3	3	0	0	0	0	0	1	2	0	0	0	0	0
	CDH23.3	62	0	0	0	0	0	18	32	11	0	0	0	0
	CDH26.7	3	0	0	0	0	0	1	2	0	0	0	0	0
Wind	WSAvg	3.0	3.6	3.4	2.9	3.4	3.0	3.0	2.7	3.0	2.6	2.6	3.1	2.9
Precipitation	PrecAvg	518	18	21	28	26	30	51	67	84	60	55	47	30
	PrecMax	696	40	57	73	100	80	90	122	163	120	94	117	101
	PrecMin	331	3	1	0	1	1	13	23	31	4	7	8	4
	PrecStd	76	9	12	17	21	16	20	24	36	29	21	26	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-8.2	-10.8	-4.2	2.2	13.7	25.6	25.9	25.0	16.3	4.9	-2.0	-10.4
		MCWB	-9.4	-11.6	-5.5	-1.5	7.3	14.9	15.7	14.6	8.7	1.7	-3.7	-11.2
	2%	DB	-17.7	-14.5	-7.5	-0.4	10.6	23.2	24.0	21.8	12.6	2.9	-4.8	-13.7
		MCWB	-18.1	-15.2	-8.9	-2.7	5.3	13.5	14.7	13.1	6.6	0.7	-5.7	-14.3
	5%	DB	-21.2	-17.2	-9.3	-2.3	7.9	21.2	22.1	19.1	10.3	1.1	-8.0	-17.4
		MCWB	-21.5	-17.6	-10.4	-3.9	3.9	12.2	13.6	11.9	5.8	-0.3	-8.7	-17.7
	10%	DB	-23.0	-19.3	-11.8	-3.9	5.8	18.8	20.1	16.6	8.3	-1.1	-12.0	-20.5
		MCWB	-23.3	-19.6	-12.8	-5.4	2.2	11.2	12.5	10.5	5.2	-2.4	-12.6	-20.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-9.6	-11.6	-5.6	-1.3	7.6	15.4	16.7	14.7	9.1	2.8	-3.5	-11.0
		MCDB	-8.4	-10.9	-4.2	0.9	13.6	24.4	24.0	22.8	14.5	4.3	-2.1	-10.1
	2%	WB	-17.9	-15.1	-8.9	-2.4	5.6	13.9	15.3	13.5	7.4	1.0	-5.7	-14.2
		MCDB	-17.7	-14.6	-7.8	-0.5	10.0	22.8	22.4	20.5	11.2	2.3	-4.7	-13.6
	5%	WB	-21.5	-17.6	-10.4	-3.7	4.0	12.6	14.2	12.4	6.4	-0.7	-8.8	-17.7
		MCDB	-21.3	-17.2	-9.3	-2.1	7.6	20.1	20.6	18.1	9.6	1.1	-8.2	-17.4
	10%	WB	-23.3	-19.6	-12.7	-5.3	2.6	11.4	13.1	11.0	5.5	-2.3	-12.5	-20.7
		MCDB	-23.1	-19.3	-11.7	-3.9	5.3	18.3	19.4	15.8	8.2	-1.1	-12.0	-20.4
Mean Daily Temperature Range	5% DB	MDBR	4.9	5.9	8.4	8.4	7.9	10.7	10.1	9.7	7.7	7.3	6.4	5.0
		MCDBR	6.9	7.5	10.2	8.4	10.7	15.1	14.4	15.5	11.5	8.5	9.7	7.2
		MCWBR	6.8	7.1	8.8	6.5	6.6	7.4	6.3	7.7	6.6	6.4	8.9	6.7
	5% WB	MCDBR	7.1	7.4	10.1	7.5	9.7	14.3	13.1	13.7	10.1	7.6	9.6	7.2
		MCWBR	6.9	7.0	8.7	5.9	6.1	7.2	6.1	7.1	5.8	5.9	8.9	6.8
Clear-Sky Solar Irradiance	taub	0.191	0.237	0.285	0.357	0.384	0.369	0.417	0.385	0.302	0.262	0.204	0.165	
	taud	2.078	2.164	2.075	1.899	1.989	2.325	2.205	2.285	2.476	2.245	2.047	1.939	
	Ebn at Noon	598	769	829	818	829	858	798	796	827	720	543	445	
	Edh at Noon	38	68	107	157	155	114	126	107	73	62	38	26	
All-Sky Solar Radiation	RadAvg	0.39	1.31	3.12	5.19	5.74	5.21	4.96	3.60	2.42	1.37	0.54	0.19	
	RadStd	0.04	0.08	0.15	0.23	0.51	0.58	0.50	0.41	0.38	0.20	0.06	0.01	

Historical Trends

	DBAvg	Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	-1.61	+1.01	+0.65	+0.57	N/A	N/A	+33

Nomenclature: See separate page